I-JEPA, anotado

Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture · M. Assran et al. (2023) · arXiv:2301.08243

Para qué son estas notas. Acompañan la semana 1 de la planificación (sesiones S1 a S3). Señalan lo esencial, lo conectan con las dos preguntas del proyecto y con *Discretizar, conjeturar, refinar*, y marcan qué dejar para después. El paper es a dos columnas: las notas de la columna izquierda van en el margen izquierdo y las de la derecha, en el derecho.

Si te quedas con cinco ideas

1. Predecir **representaciones** y no píxeles es el ingrediente principal (tabla 7: 66.9 frente a 40.7).
2. La máscara es el sesgo inductivo: objetivos medianos (15–20 % del área), varios (4) y un contexto amplio (tablas 6 y 8–10). Esa escala la elige el diseñador, no el modelo.
3. Lo abstracto conserva forma, pose y posición, y descarta textura y detalle fino (figuras 6–8). Por eso MAE, que reconstruye píxeles, todavía gana en contar objetos (tabla 4).
4. Un predictor estrecho, es decir, un cuello de botella, rinde mejor que uno ancho (tabla 14): restringir capacidad ayuda sin discretizar nada.
5. El aporte de I-JEPA a nuestras preguntas está en P1 (lo que se discretiza es **la tarea**) y en C3 (la escala). Sobre P2 casi no dice nada.

Ruta por sesión. S1: págs. 1–4. S2: págs. 5–8 y 14–15. S3: págs. 7–8, 12–13 y 16–17. Las páginas 9–11 son referencias; solo se marcan las que vale la pena buscar.

Colores de las notas

CLAVE	Idea esencial: lo que conviene recordar del paper.
PREGUNTA	Conexión con P1, P2, los mecanismos M1–M6 o las preguntas C1–C10 del .epub.
PROFUNDIZAR	Para más adelante: lecturas, experimentos que faltan, extensiones posibles.
CUIDADO	Límites de la evidencia, comparaciones dudosas, erratas.
APUNTE	Concepto previo o notación que el paper da por sabida.
EJERCICIO	Algo que conviene verificar a mano.
EXTRA	Detalle que se puede saltar en una primera lectura.

Cómo leer. Cada nota tiene un número. El mismo número aparece junto a una **barra de color** al lado del pasaje del original, y una línea fina las une. Si la nota se refiere a una ecuación o figura precisa, esta aparece **enmarcada** con el mismo color.

Etiquetas usadas en las notas

Remiten a *Discretizar, conjeturar, refinar* (abreviado **ep.**), y las sesiones a la planificación de lectura.

P1	¿Por qué, al discretizar, mejoran los algoritmos? ¿Qué principios son comunes?
P2	¿Cómo se modelan las reglas? ¿Qué genera las hipótesis? ¿Cuándo refinar y cuándo ampliar?
Mecanismos (ep. §3)	
M1	Compresión y navaja de Occam (§3.1)
M2	Corrección de errores y estabilidad (§3.2)
M3	Multimodalidad: elegir en vez de promediar (§3.3)
M4	Combinatoria sin mezcla, principio particulado (§3.4)
M5	Búsqueda y enumeración (§3.5)
M6	Comunicación y transmisión (§3.6)
§3.7	Lo que se pierde al discretizar
Preguntas concretas (ep. §7)	
C1	¿Discretización, cuello de botella o largo variable?
C2	¿Qué mecanismo actúa en cada caso?
C3	¿Dónde está el nivel básico?
C4	¿Cuántos factores necesita un mundo?
C5	¿Es un factor una regla?
C6	¿De dónde salen las hipótesis en cada modelo?
C7	¿Cuándo pasar de refinar a reemplazar?
C8	Varias tareas: ¿reutilizar o crear?
C9	¿Qué parte es innata?
C10	¿Qué hace el sueño con las reglas?

Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture

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Abstract

This paper demonstrates an approach for learning highly semantic image representations without relying on hand-crafted data-augmentations. We introduce the Image-based Joint-Embedding Predictive Architecture (I-JEPA), a non-generative approach for self-supervised learning from images. The idea behind I-JEPA is simple: from a single context block, predict the representations of various target blocks in the same image. A core design choice to guide I-JEPA towards producing semantic representations is the masking strategy; specifically, it is crucial to (a) sample target blocks with sufficiently large scale (semantic), and to (b) use a sufficiently informative (spatially distributed) context block. Empirically, when combined with Vision Transformers, we find I-JEPA to be highly scalable. For instance, we train a ViT-Huge/14 on ImageNet using 16 A100 GPUs in under 72 hours to achieve strong downstream performance across a wide range of tasks, from linear classification to object counting and depth prediction.

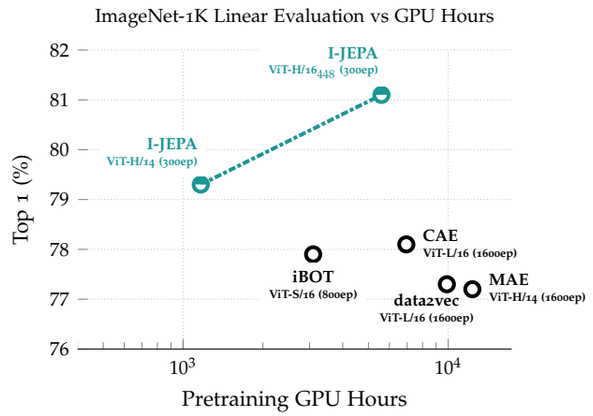


Figure 1. **ImageNet Linear Evaluation**. The I-JEPA method learns semantic image representations without using any view data augmentations during pretraining. By predicting in representation space, I-JEPA produces semantic representations while using less compute than previous methods.

1. Introduction

In computer vision, there are two common families of approaches for self-supervised learning from images: invariance-based methods [1, 4, 10, 17, 18, 24, 35, 37, 74] and generative methods [8, 28, 36, 57].

Invariance-based pretraining methods optimize an encoder to produce similar embeddings for two or more views of the same image [15, 20], with image views typically constructed using a set of hand-crafted data augmentations, such as random scaling, cropping, and color jittering [20], amongst others [35]. These pretraining methods can produce representations of a high semantic level [4, 18], but they also introduce strong biases that may be detrimental for certain downstream tasks or even for pretraining tasks with different data distributions [2]. Often, it is unclear how to generalize these biases for tasks requiring different levels of abstraction. For example, image classification and instance segmentation do not require the same invariances [11]. Additionally, it is not straightforward to generalize these image-specific augmentations to other modalities such as audio.

Cognitive learning theories have suggested that a driving mechanism behind representation learning in biological systems is the adaptation of an internal model to predict sensory input responses [31, 59]. This idea is at the core of self-supervised generative methods, which remove or corrupt portions of the input and learn to predict the corrupted content [9, 36, 57, 67, 68, 71]. In particular, mask-denoising approaches learn representations by reconstructing randomly masked patches from an input, either at the pixel or token level. Masked pretraining tasks require less prior knowledge than view-invariance approaches and easily generalize beyond the image modality [8]. However, the

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1 **CLAVE** **Tesis:** desde un bloque de contexto, predecir la **representación** (no los píxeles) de otros bloques de la misma imagen, sin aumentos de datos hechos a mano.

2 **CLAVE** Las dos condiciones del método: objetivos **grandes** (semánticos) y un contexto **informativo** y repartido en el espacio. Ambas son decisiones de diseño (tablas 6 y 8–10).

3 **PREGUNTA** La escala de los objetivos la fija el diseñador. Es la pista para C3 y para la pregunta "¿quién decide los cortes?" (ep. tabla 2, fila "la tarea").

4 **CUIDADO** La figura 1 mide eficiencia (horas-GPU), no solo exactitud. El empate con iBOT (81.1 frente a 81.0, tabla 1) exige un modelo más grande y resolución 448.

5 **PREGUNTA** "Tareas que requieren distintos niveles de abstracción": cada tarea pide una granularidad distinta. Es la idea del nivel básico de Rosch (ep. §3.8, C3).

6 **APUNTE** Dos familias previas. **Invarianza** (DINO, iBOT, SimCLR): dos versiones aumentadas de una imagen deben dar el mismo embedding. **Generativas** (MAE): reconstruir los píxeles ocultos.

7 **APUNTE** Las refs. 31 y 59 son Friston y Rao y Ballard: codificación predictiva. Percibir como predecir, en la línea de Helmholtz y Craik (ep. §2.5). JEPA es esa idea en un espacio abstracto.

8 **CLAVE** Crítica a la invarianza: recortes, cambios de color, etc. son **sesgos inductivos hechos a mano**. Sirven para clasificar, pero pueden dañar tareas que necesitan otra invarianza, como contar o segmentar.

9 **PREGUNTA** Los aumentos son conocimiento "de fábrica" (ep. §4.3, Kant y Spelke). I-JEPA no elimina el sesgo: lo **traslada** al diseño de la máscara (C9).

10 **CLAVE** Ventaja de enmascarar: pide menos conocimiento previo que la invarianza y sirve en cualquier modalidad. Por eso JEPA-Anything puede aplicarlo a células, clima o moléculas.

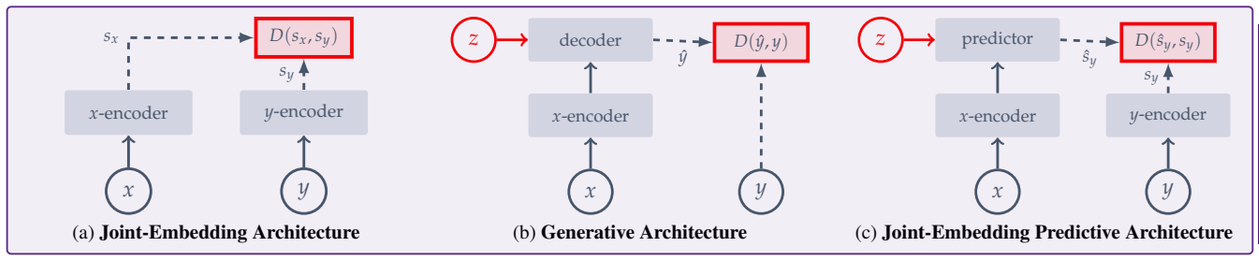


Figure 2. Common architectures for self-supervised learning, in which the system learns to capture the relationships between its inputs. The objective is to assign a high energy (large scaler value) to incompatible inputs, and to assign a low energy (low scaler value) to compatible inputs. **(a)** Joint-Embedding Architectures learn to output similar embeddings for compatible inputs x, y and dissimilar embeddings for incompatible inputs. **(b)** Generative Architectures learn to directly reconstruct a signal y from a compatible signal x, using a decoder network that is conditioned on additional (possibly latent) variables z to facilitate reconstruction. **(c)** Joint-Embedding Predictive Architectures learn to predict the embeddings of a signal y from a compatible signal x, using a predictor network that is conditioned on additional (possibly latent) variables z to facilitate prediction.

resulting representations are typically of a lower semantic level and underperform invariance-based pretraining in off-the-shelf evaluations (e.g., linear-probing) and in transfer settings with limited supervision for semantic classification tasks [4]. Consequently, a more involved adaptation mechanism (e.g., end-to-end fine-tuning) is required to reap the full advantage of these methods.

In this work, we explore how to improve the semantic level of self-supervised representations without using extra prior knowledge encoded through image transformations. To that end, we introduce a joint-embedding predictive architecture [48] for images (I-JEPA). An illustration of the method is provided in Figure 3. The idea behind I-JEPA is to predict missing information in an abstract representation space; e.g., given a single context block, predict the representations of various target blocks in the same image, where target representations are computed by a learned target-encoder network.

Compared to generative methods that predict in pixel/token space, I-JEPA makes use of abstract prediction targets for which unnecessary pixel-level details are potentially eliminated, thereby leading the model to learn more semantic features. Another core design choice to guide I-JEPA towards producing semantic representations is the proposed multi-block masking strategy. Specifically, we demonstrate the importance of predicting sufficiently large target blocks in the image, using an informative (spatially distributed) context block.

Through an extensive empirical evaluation, we demonstrate that:

- I-JEPA learns strong off-the-shelf representations without the use of hand-crafted view augmentations (cf. Fig.1). I-JEPA outperforms pixel-reconstruction methods such as MAE [36] on ImageNet-1K linear probing, semi-supervised 1% ImageNet-1K, and semantic transfer tasks.
- I-JEPA is competitive with view-invariant pretraining approaches on semantic tasks and achieves better performance on low-level visions tasks such as object counting and depth prediction (Sections 5 and 6). By using a simpler model with less rigid inductive bias, I-JEPA is applicable to a wider set of tasks.
- I-JEPA is also scalable and efficient (Section 7). Pretraining a ViT-H/14 on ImageNet requires less than 1200 GPU hours, which is over $2.5\times$ faster than a ViT-S/16 pretrained with iBOT [79] and over $10\times$ more efficient than a ViT-H/14 pretrained with MAE. Predicting in representation space significantly reduces the total computation needed for self-supervised pretraining.

2. Background

Self-supervised learning is an approach to representation learning in which a system learns to capture the relationships between its inputs. This objective can be readily described using the framework of Energy-Based Models (EBMs) [49] in which the self-supervised objective is to assign a high energy to incompatible inputs, and to assign a low energy to compatible inputs. Many existing generative and non-generative approaches to self-supervised learning can indeed be cast in this framework; see Figure 2.

Joint-Embedding Architectures. Invariance-based pretraining can be cast in the framework of EBMs using a Joint-Embedding Architecture (JEA), which learns to output similar embeddings for compatible inputs, $\boldsymbol{x}, \boldsymbol{y}$, and dissimilar embeddings for incompatible inputs; see Figure 2a. In the context of image-based pretraining, compatible $\boldsymbol{x}, \boldsymbol{y}$ pairs are typically constructed by randomly applying hand-crafted data augmentations to the same input image [20].

The main challenge with JEAs is representation collapse, wherein the energy landscape is flat (i.e., the encoder produces a constant output regardless of the input). During the past few years, several approaches have been investi-

1 **CLAVE** **La figura más importante del paper.** (a) JEA: embeddings parecidos para entradas compatibles; riesgo de colapso. (b) Generativa: reconstruir y en el espacio de entrada. (c) JEPA: predecir el **embedding** de y desde el de x, condicionado en z.

2 **PREGUNTA** Pregunta de S1: ¿qué decide **no explicar** cada una? (a) Todo lo que cambian los aumentos. (b) Nada: debe explicar cada píxel. (c) Lo que el target encoder descarte. Conecta con M1 y M3.

3 **CUIDADO** "Mejor en tareas de bajo nivel" se dice **respecto de los métodos de invarianza**. En la tabla 4, MAE (píxeles) gana en Clevr/Count: 90.5 frente a 86.7. Lo abstracto pierde detalle (ep. §3.7).

4 **APUNTE** **Linear probing**: congelar el encoder y entrenar solo un clasificador lineal encima. Mide qué tan "legible" es la representación sin adaptarla. El **fine-tuning**, en cambio, la ajusta entera.

5 **EXTRA** Eficiencia: ViT-H/14 en menos de 1200 horas-GPU, unas 10 veces menos que MAE. Es un argumento práctico más que conceptual.

6 **CLAVE** La apuesta: el objetivo es abstracto porque lo produce un encoder **aprendido**, y los detalles de píxel innecesarios "se eliminan potencialmente". Fíjate en el "potencialmente": no está garantizado.

7 **APUNTE** **Modelos basados en energía** (LeCun et al., 2006): la energía mide incompatibilidad. Aprender es bajar la energía de los pares compatibles y evitar que baje para los incompatibles.

8 **PREGUNTA** ¿Por qué ayuda? Hay dos lecturas posibles en el ep.: M1 (el objetivo lleva menos información que el píxel) o M3 (queda menos incertidumbre que promediar). El paper no separa ambas.

9 **APUNTE** **Colapso**: el encoder entrega la misma salida para todo, la energía queda plana y la pérdida es mínima sin aprender nada. Todo método no generativo necesita un mecanismo anticolapso.

gated to prevent representation collapse, such as contrastive losses that explicitly push apart embeddings of negative examples [15,24,37], non-contrastive losses that minimize the informational redundancy across embeddings [10, 74], and clustering-based approaches that maximize the entropy of the average embedding [4, 5, 18]. There are also heuristic approaches that leverage an asymmetric architectural design between the x-encoder and y-encoder to avoid collapse [8, 24, 35].

Generative Architectures. Reconstruction-based methods for self-supervised learning can also be cast in the framework of EBMs using Generative Architectures; see Figure 2b. Generative Architectures learn to directly reconstruct a signal $\boldsymbol{y}$ from a compatible signal $\boldsymbol{x}$, using a decoder network that is conditioned on an additional (possibly latent) variable $\boldsymbol{z}$ to facilitate reconstruction. In the context of image-based pretraining, one common approach in computer vision is to produce compatible $\boldsymbol{x}, \boldsymbol{y}$ pairs using masking [9, 38] where $\boldsymbol{x}$ is a copy of the image $\boldsymbol{y}$, but with some of the patches masked. The conditioning variable $\boldsymbol{z}$ then corresponds to a set of (possibly learnable) mask and position tokens, that specifies to the decoder which image patches to reconstruct. Representation collapse is not a concern with these architectures as long as the informational capacity of $\boldsymbol{z}$ is low compared to the signal $\boldsymbol{y}$.

Joint-Embedding Predictive Architectures. As shown in Figure 2c, Joint-Embedding Predictive Architectures [48] are conceptually similar to Generative Architectures; however, a key difference is that the loss function is applied in embedding space, not input space. JEPAs learn to predict the embeddings of a signal $\boldsymbol{y}$ from a compatible signal $\boldsymbol{x}$, using a predictor network that is conditioned on an additional (possibly latent) variable $\boldsymbol{z}$ to facilitate prediction. Our proposed I-JEPA provides an instantiation of this architecture in the context of images using masking; see Figure 3.

In contrast to Joint-Embedding Architectures, JEPAs do not seek representations invariant to a set of hand-crafted data augmentations, but instead seek representations that are predictive of each other when conditioned on additional information $\boldsymbol{z}$. However, as with Joint-Embedding Architectures, representation collapse is also a concern with JEPAs; we leverage an asymmetric architecture between the $\boldsymbol{x}$- and $\boldsymbol{y}$-encoders to avoid representation collapse.

3. Method

We now describe the proposed Image-based Joint-Embedding Predictive Architecture (I-JEPA), illustrated in Figure 3. The overall objective is as follows: given a context block, predict the representations of various target blocks

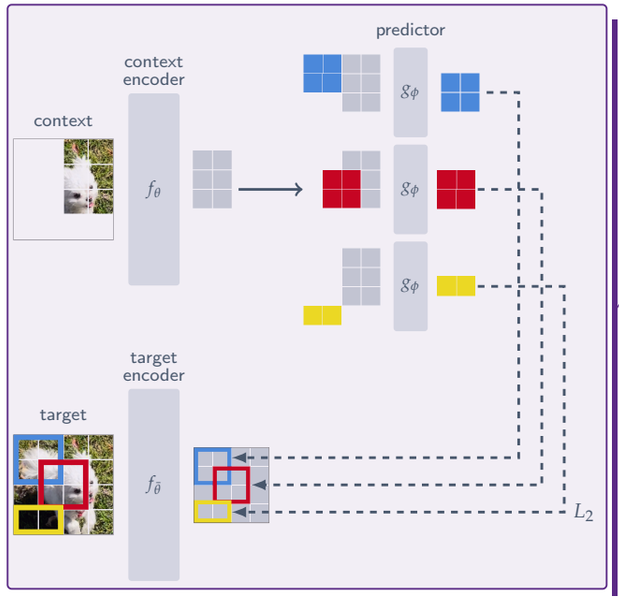


Figure 3. **I-JEPA.** The Image-based Joint-Embedding Predictive Architecture uses a single context block to predict the representations of various target blocks originating from the same image. The context encoder is a Vision Transformer (ViT), which only processes the visible context patches. The predictor is a narrow ViT that takes the context encoder output and, conditioned on positional tokens (shown in color), predicts the representations of a target block at a specific location. The target representations correspond to the outputs of the target-encoder, the weights of which are updated at each iteration via an exponential moving average of the context encoder weights.

in the same image. We use a Vision Transformer [29, 63] (ViT) architecture for the context-encoder, target-encoder, and predictor. A ViT is composed of a stack of transformer layers, each consisting of a self-attention [66] operation followed by a fully-connected MLP. Our encoder/predictor architecture is reminiscent of the generative masked autoencoders (MAE) [36] method. However, one key difference is that the I-JEPA method is non-generative and the predictions are made in representation space.

Targets. We first describe how we produce the targets in the I-JEPA framework: in I-JEPA, the targets correspond to the representations of image blocks. Given an input image $\boldsymbol{y}$, we convert it into a sequence of N non-overlapping patches, and feed this through the target-encoder $f_{\bar{\theta}}$ to obtain a corresponding patch-level representation $\boldsymbol{s}_y = \{\boldsymbol{s}_{y_1}, \ldots, \boldsymbol{s}_{y_N}\}$ where $\boldsymbol{s}_{y_k}$ is the representation associated with the kth patch. To obtain the targets for our loss, we randomly sample M (possibly overlapping) blocks from the target representations $\boldsymbol{s}_y$. We denote by B_i the mask corresponding of the ith block and by $\boldsymbol{s}_y(i) = \{\boldsymbol{s}_{y_j}\}_{j \in B_i}$ its

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1 **PROFUNDIZAR** Formas de evitar el colapso: contrastiva (negativos), reducción de redundancia (VICReg, Barlow Twins: ep. §3.1, Barlow 1961), **clustering** o una arquitectura asimétrica (EMA). JEPA-Anything usará **además** un término de varianza tipo VICReg.

2 **CLAVE** Tres redes. **Context encoder** f_θ: solo ve los parches visibles. **Target encoder** $f_{\bar{\theta}}$: ve la imagen completa y es un promedio exponencial (EMA) del primero. **Predictor** g_φ: un ViT estrecho condicionado en posiciones.

3 **APUNTE** El contexto solo procesa los parches visibles: es barato. El target encoder procesa la imagen completa, pero sin gradiente.

4 **PREGUNTA** Idea fina: en lo generativo no hay colapso **si** z **lleva poca información** comparada con y. Es un argumento de cuello de botella (M1): limitar z obliga a usar x.

5 **APUNTE** En I-JEPA, z son los **tokens de posición** de los bloques objetivo. Le dicen al predictor **dónde** predecir, nunca **qué**.

6 **CUIDADO** La prevención del colapso es **empírica** (EMA y predictor). El paper no da garantías; solo "una arquitectura asimétrica".

7 **CLAVE** Los objetivos se recortan de la **salida** del target encoder, que vio la imagen entera: cada parche objetivo trae contexto global. Por eso es "semántico" (tabla 11: 67.3 frente a 56.1).

8 **PROFUNDIZAR** Para entender por qué no colapsa: BYOL (Grill et al., 2020) y la ref. 62, Tian, Chen y Ganguli (2021), sobre la dinámica del aprendizaje sin pares negativos.



Figure 4. Examples of our context and target-masking strategy. Given an image, we randomly sample 4 target blocks with scale in the range $(0.15, 0.2)$ and aspect ratio in the range $(0.75, 1.5)$. Next, we randomly sample a context block with scale in the range $(0.85, 1.0)$ and remove any overlapping target blocks. Under this strategy, the target-blocks are relatively semantic, and the context-block is informative, yet sparse (efficient to process).

patch-level representation. Typically, we set M equal to 4, and sample the blocks with a random aspect ratio in the range $(0.75, 1.5)$ and random scale in the range $(0.15, 0.2)$. Note that the target blocks are obtained by masking the *output* of the target-encoder, not the input. This distinction is crucial to ensure target representations of a high semantic level; see, e.g., [8].

Context. Recall, the goal behind I-JEPA is to predict the target block representations from a single context block. To obtain the context in I-JEPA, we first sample a single block $\boldsymbol{x}$ from the image with a random scale in the range $(0.85, 1.0)$ and unit aspect ratio. We denote by B_x the mask associated with the context block $\boldsymbol{x}$. Since the target blocks are sampled independently from the context block, there may be significant overlap. To ensure a non-trivial prediction task, we remove any overlapping regions from the context block. Figure 4 shows examples of various context and target blocks in practice. Next, the masked context block, $\boldsymbol{x}$, is fed through the context encoder f_θ to obtain a corresponding patch-level representation $\boldsymbol{s}_x = \{\boldsymbol{s}_{x_j}\}_{j \in B_x}$.

Prediction. Given the output of the context encoder, $\boldsymbol{s}_x$, we wish to predict the M target block representations $\boldsymbol{s}_y(1), \ldots, \boldsymbol{s}_y(M)$. To that end, for a given target block $\boldsymbol{s}_y(i)$ corresponding to a target mask B_i, the predictor $g_\phi(\cdot, \cdot)$ takes as input the output of the context encoder $\boldsymbol{s}_x$ and a mask token for each patch we wish to predict, $\{\boldsymbol{m}_j\}_{j \in B_i}$, and outputs a patch-level prediction $\hat{\boldsymbol{s}}_y(i) = \{\hat{\boldsymbol{s}}_{y_j}\}_{j \in B_i} = g_\phi(\boldsymbol{s}_x, \{\boldsymbol{m}_j\}_{j \in B_i})$. The mask tokens are parameterized by a shared learnable vector with an added positional embedding. Since we wish to make predictions for M target blocks, we apply our predictor M times, each time conditioning on the mask tokens corresponding to the target-block locations we wish to predict, and obtain predictions $\hat{\boldsymbol{s}}_y(1), \ldots, \hat{\boldsymbol{s}}_y(M)$.

Loss. The loss is simply the average L_2 distance between the predicted patch-level representations $\hat{\boldsymbol{s}}_y(i)$ and the target patch-level representation $\boldsymbol{s}_y(i)$; i.e.,

$$\frac{1}{M} \sum_{i=1}^{M} D\left(\hat{\boldsymbol{s}}_y(i), \boldsymbol{s}_y(i)\right) = \frac{1}{M} \sum_{i=1}^{M} \sum_{j \in B_i} \|\hat{\boldsymbol{s}}_{y_j} - \boldsymbol{s}_{y_j}\|_2^2.$$

The parameters of the predictor, ϕ, and context encoder, θ, are learned through gradient-based optimization, while the parameters of the target encoder $\bar{\theta}$ are updated via an exponential moving average of the context-encoder parameters. The use of an exponential moving average target-encoder has proven essential for training JEAs with Vision Transformers [18, 25, 79], we find the same to be true for I-JEPA.

4. Related Work

A long line of work has explored visual representation learning by predicting the values of missing or corrupted sensory inputs. Denoising autoencoders use random noise as input corruption [67]. Context encoders regress an entire image region based on its surrounding [57]. Other works cast image colorization as a denoising task [46, 47, 77].

The idea of image denoising has recently been revisited in the context of masked image modelling [9, 36, 71], where a Vision Transformer [29] is used to reconstruct missing input patches. The work on Masked Autoencoders (MAE) [36] proposed an efficient architecture that only requires the encoder to process visible image patches. By reconstructing missing patches in pixels space, MAE achieves strong performance when fine-tuned end-to-end on large labeled datasets and exhibits good scaling properties. BEiT [9] predicts the value of missing patches in a tokenized space; specifically, tokenizing image patches using a frozen discreteVAE, which is trained on a dataset containing 250 million images [58]. Yet, pixel-level pre-training has been shown to outperform BEiT for fine-tuning [36]. Another work, SimMIM [71], explores reconstruction targets based on the classic Histogram of Gradients [27] feature space, and demonstrates some advantage over pixel space reconstruction. Different from those works, our representation space is learned during training through a Joint-Embedding Predictive Architecture. Our goal is to learn semantic representations that do not require extensive fine-tuning on downstream tasks.

Closest to our work is data2vec [8] and Context Autoencoders [25]. The data2vec method learns to predict the rep-

4 CLAVE **Receta de la máscara.** 4 bloques objetivo de escala 15–20 % del área y razón de aspecto 0.75–1.5. Un contexto de 85–100 % **sin** las zonas objetivo. Estos números son el sesgo inductivo del método.

5 PREGUNTA Las escalas no salen del modelo: se buscaron a mano (tabla 8). ¿Podría el modelo inferir su propia escala, como el nivel básico de Rosch? (C3).

7 APUNTE Se quita del contexto lo que se va a predecir. Si no, la tarea sería copiar.

9 APUNTE El predictor recibe s_x y un **mask token** por parche objetivo: un vector aprendido compartido más el embedding posicional. Se aplica M veces, una por bloque.

10 PREGUNTA Hay **una sola** conjetura determinista por bloque: no existe una variable latente para muestrear alternativas. Es lo opuesto a generar hipótesis por muestreo (Dasgupta et al., ep. §4.3; C6).

1 CLAVE **Pérdida (1)**: promedio sobre los M bloques de $\sum_{j \in B_i} \left\| \hat{s}_{y_j} - s_{y_j} \right\|_2^2$. Es una distancia L2 **entre representaciones**, nunca entre píxeles.

2 PREGUNTA L2 con un predictor determinista ⇒ se predice la **media condicional** del embedding. Si el contenido oculto es ambiguo, se promedia. I-JEPA lo esquiva porque en espacio abstracto queda menos incertidumbre (ep. §3.3, M3).

3 APUNTE EMA: $\bar{\theta} \leftarrow m\bar{\theta} + (1-m)\theta$, con m de 0.996 a 1.0 (apéndice A). El target encoder no recibe gradiente; esta asimetría, junto con el predictor, evita el colapso en la práctica.

6 EXTRA §4 se lee rápido. El pariente más cercano es data2vec (también predice representaciones de parches ocultos); la diferencia práctica es la máscara multibloque y el costo.

8 PREGUNTA BEiT discretiza **el objetivo**: tokens de un dVAE, pariente de VQ-VAE. Aquí discretizar no ganó: MAE con píxeles lo supera en fine-tuning. Dato para P1: discretizar el objetivo, por sí solo, no basta.

Method	Arch.	Epochs	Top-1
Methods without view data augmentations			
data2vec [8]	ViT-L/16	1600	77.3
MAE [36]	ViT-B/16	1600	68.0
	ViT-L/16	1600	76.0
	ViT-H/14	1600	77.2
CAE [22]	ViT-B/16	1600	70.4
	ViT-L/16	1600	78.1
I-JEPA	ViT-B/16	600	72.9
	ViT-L/16	600	77.5
	ViT-H/14	300	79.3
	ViT-H/16_{448}	300	**81.1**
Methods using extra view data augmentations			
SimCLR v2 [21]	RN152 (2×)	800	79.1
DINO [18]	ViT-B/8	300	80.1
iBOT [79]	ViT-L/16	250	**81.0**

Table 1. **ImageNet**. Linear-evaluation on ImageNet-1k (the ViT-H/16_{448} is pretrained at at a resolution of 448×448). I-JEPA improves linear probing performance compared to other methods that do not rely on hand-crafted view data-augmentations during pretraining. Moreover, I-JEPA demonstrates good scalability — the larger I-JEPA model matches the performance of view-invariance approaches without requiring view data-augmentations.

Method	Arch.	Epochs	Top-1
Methods without view data augmentations			
data2vec [8]	ViT-L/16	1600	73.3
MAE [36]	ViT-L/16	1600	67.1
	ViT-H/14	1600	71.5
I-JEPA	ViT-L/16	600	69.4
	ViT-H/14	300	73.3
	ViT-H/16_{448}	300	**77.3**
Methods using extra view data augmentations			
iBOT [79]	ViT-B/16	400	69.7
DINO [18]	ViT-B/8	300	70.0
SimCLR v2 [35]	RN151 (2×)	800	70.2
BYOL [35]	RN200 (2×)	800	71.2
MSN [4]	ViT-B/4	300	**75.7**

Table 2. **ImageNet-1%**. Semi-supervised evaluation on ImageNet-1K using only 1% of the available labels. Models are adapted via fine-tuning or linear-probing, depending on whichever works best for each respective method. ViT-H/16_{448} is pretrained at at a resolution of 448×448. I-JEPA pretraining outperforms MAE which also does not rely on hand-crafted data-augmentations during pretraining. Moreover, I-JEPA benefits from scale. A ViT-H/16 trained at resolution 448 surpasses previous methods including methods that leverage extra hand-crafted data-augmentations.

resentation of missing patches computed through an online target encoder; by avoiding handcrafted augmentations, the method can be applied to diverse modalities with promising results in vision, text and speech. Context Autoencoders use an encoder/decoder architecture optimized via the sum of a reconstruction loss and an alignment constraint, which enforces predictability of missing patches in representation space. Compared to these methods, I-JEPA exhibits significant improvements in computational efficiency and learns more semantic off-the-shelf representations. Concurrent to our work, data2vec-v2 [7] explores efficient architectures for learning with various modalities.

We also compare I-JEPA with various methods based on joint-embedding architectures; e.g., DINO [18], MSN [4] and iBOT [79]. Theses methods rely on hand-crafted data augmentations during pretraining to learn semantic image representations. The work on MSN [4], uses masking as an additional data-augmentation during pretraining, while iBOT combines a data2vec-style patch-level reconstruction loss with the DINO view-invariance loss. Common to these approaches is the need to process multiple user-generated views of each input image, thereby hindering scalability. By contrast, I-JEPA only requires processing a single view of each image. We find that a ViT-Huge/14 trained with I-JEPA requires less computational effort than a ViT-Small/16 trained with iBOT.

5. Image Classification

To demonstrate that I-JEPA learns high-level representations without relying on hand-crafted data-augmentations, we report results on various image classification tasks using the linear probing and partial fine-tuning protocols. In this section, we consider self-supervised models that have been pretrained on the ImageNet-1K dataset [60]. Pretraining and evaluation implementation details are described in the Appendix A. All I-JEPA models are trained at resolution 224×224 pixels, unless stated otherwise.

ImageNet-1K. Table 1 shows performance on the common ImageNet-1K linear-evaluation benchmark. After self-supervised pretraining, the model weights are frozen and a linear classifier is trained on top using the full ImageNet-1K training set. Compared to popular methods such as Masked Autoencoders (MAE) [36], Context Autoencoders (CAE) [22], and data2vec [8], which also do not rely on extensive hand-crafted data-augmentations during pretraining, we see that I-JEPA significantly improves linear probing performance, while using less computational effort (see section 7). By leveraging the improved efficiency of I-JEPA, we can train larger models that outperform the best CAE model while using a fraction of the compute. I-JEPA also benefits from scale; in particular, a ViT-H/16 trained at resolution 448×448 pixels matches the performance of view-

2 CUIDADO Tabla 1. Sin aumentos, I-JEPA gana claramente: 79.3 frente a 77.2 de MAE con ViT-H/14, y con 300 épocas en vez de 1600. Frente a métodos **con** aumentos, solo empata usando ViT-H a 448 px (81.1 frente a 81.0 de iBOT).

1 PREGUNTA Tabla 2: 1 % de las etiquetas, unas 12 imágenes por clase. Es el régimen de pocos ejemplos del juego de los números (ep. §4.1): ¿qué hace que una representación generalice con tan pocos datos?

4 EXTRA Comparación de costo: DINO, MSN e iBOT procesan varias vistas por imagen; I-JEPA, una sola. Se puede pasar rápido.

3 EXTRA §5 en lectura rápida (S2): basta con retener **qué** se afirma.

Method	Arch.	CIFAR100	Places205	iNat18
Methods without view data augmentations				
data2vec [8]	ViT-L/16	81.6	54.6	28.1
MAE [36]	ViT-H/14	77.3	55.0	32.9
I-JEPA	ViT-H/14	**87.5**	**58.4**	**47.6**
Methods using extra view data augmentations				
DINO [18]	ViT-B/8	84.9	57.9	55.9
iBOT [79]	ViT-L/16	**88.3**	**60.4**	**57.3**

Table 3. **Linear-probe transfer for image classification**. Linear-evaluation on downstream image classification tasks. I-JEPA significantly outperforms previous methods that also do not use augmentations (MAE and data2vec), and decreases the gap with the best view-invariance-based methods that leverage hand-crafted data augmentations during pretraining.

Method	Arch.	Clevr/Count	Clevr/Dist
Methods without view data augmentations			
data2vec [8]	ViT-L/16	85.3	71.3
MAE [36]	ViT-H/14	**90.5**	**72.4**
I-JEPA	ViT-H/14	86.7	**72.4**
Methods using extra data augmentations			
DINO [18]	ViT-B/8	86.6	53.4
iBOT [79]	ViT-L/16	85.7	62.8

Table 4. **Linear-probe transfer for low-level tasks**. Linear-evaluation on downstream low-level tasks consisting of object counting (Clevr/Count) and depth prediction (Clevr/Dist). The I-JEPA method effectively captures low-level image features during pretraining and outperforms view-invariance based methods on tasks such object counting and depth prediction.

invariant approaches such as iBOT [79], despite avoiding the use of hand-crafted data-augmentations.

Low-Shot ImageNet-1K. Table 2 shows performance on the 1% ImageNet benchmark. Here the idea is to adapt the pretrained models for ImageNet classification using only 1% of the available ImageNet labels, corresponding to roughly 12 or 13 images per class. Models are adapted via fine-tuning or linear-probing, depending on whichever works best for each respective method. I-JEPA outperforms MAE while requiring less pretraining epochs when using a similar encoder architecture. I-JEPA, using a ViT-H/14 architecture, matches the performance of a ViT-L/16 pretrained with data2vec [8], while using significantly less computational effort (see Section 7). By increasing the image input resolution, I-JEPA outperforms previous methods including joint-embedding methods that do leverage extra hand-crafted data-augmentations during pretraining, such as MSN [4], DINO [17], and iBOT [79].

Transfer learning. Table 3 shows performance on various downstream image classification tasks using a linear probe. I-JEPA significantly outperforms previous methods that do not use augmentations (MAE and data2vec), and decreases the gap with the best view-invariance-based methods, which leverage hand-crafted data augmentations during pretraining, even surpassing the popular DINO [18] on CIFAR100 and Place205 with a linear probe.

6. Local Prediction Tasks

As demonstrated in Section 5, I-JEPA learns semantic image representations that significantly improve the downstream image classification performance of previous methods, such as MAE and data2vec. Additionally, I-JEPA benefits from scale and can close the gap, and even surpass, view-invariance based methods that leverage extra hand-crafted data augmentations. In this section, we find that I-JEPA also learns local image features and surpasses view-invariance based methods on low-level and dense prediction tasks, such as object counting and depth prediction.

Table 4 shows performance on various low-level tasks using a linear probe. After pretraining, the encoder weights are frozen and a linear model is trained on top to perform object-counting and depth prediction on the Clevr dataset [43]. Compared to view-invariance methods such as DINO and iBOT, the I-JEPA method effectively captures low-level image features during pretraining and outperforms them in object counting (Clevr/Count) and (by a large margin) depth prediction (Clevr/Dist).

7. Scalability

Model Efficiency. I-JEPA is highly scalable compared to previous approaches. Figure 5 shows semi-supervised evaluation on 1% ImageNet-1K as a function of GPU hours. I-JEPA requires less compute than previous methods and achieves strong performance without relying on hand-crafted data-augmentations. Compared to reconstruction-based methods, such as MAE, which directly use pixels as targets, I-JEPA introduces extra overhead by computing targets in representation space (about 7% slower time per iteration). However, since I-JEPA converges in roughly $5\times$ fewer iterations, we still see significant compute savings in practice. Compared to view-invariance based methods, such as iBOT, which rely on hand-crafted data augmentations to create and process multiple views of each image, I-JEPA also runs significantly faster. In particular, a huge I-JEPA model (ViT-H/14) requires less compute than a small iBOT model (ViT-S/16).

Scaling data size. We also find I-JEPA to benefit from pretraining with larger datasets. Table 5 shows transfer

2 **PREGUNTA** En iNat18 (especies, categorías muy finas), I-JEPA logra 47.6 frente a 57.3 de iBOT: la brecha mayor está en lo más fino. ¿Pierde la abstracción el nivel subordinado ("labrador" frente a "perro")? Es una hipótesis para C3, no un resultado.

3 **CLAVE** Tabla 3: I-JEPA supera a los métodos sin aumentos y achica la distancia con los que sí los usan. En CIFAR100 y Places205 incluso supera a DINO.

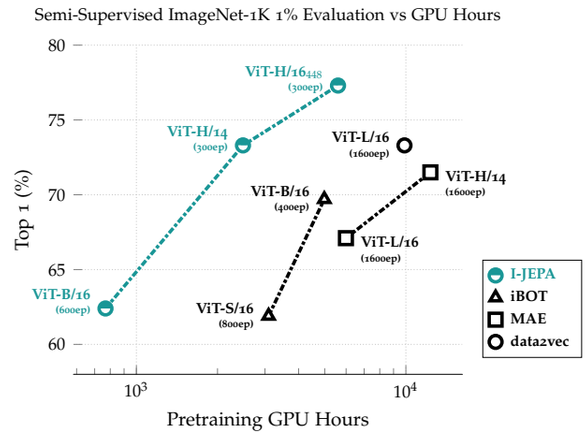
1 **CUIDADO** Tabla 4. Clevr/Count: MAE 90.5, I-JEPA 86.7. Clevr/Dist: empate en 72.4. I-JEPA gana a DINO e iBOT, no a MAE. Contar y medir distancias exige detalle que lo abstracto puede descartar (ep. §3.7).

4 **EXTRA** §7: 7 % más lento por iteración, pero converge en unas 5 veces menos iteraciones.

Pretrain	Arch.	CIFAR100	Place205	INat18	Clevr/Count	Clevr/Dist
IN1k	ViT-H/14	87.5	58.4	47.6	86.7	72.4
IN22k	ViT-H/14	**89.5**	57.8	50.5	**88.6**	**75.0**
IN22k	ViT-G/16	**89.5**	**59.1**	**55.3**	86.7	73.0

Table 5. **Ablating dataset and model size**. Evaluating impact of pre-training dataset size and model size on transfer tasks. I-JEPA benefits from larger more diverse datasets. When increasing the size of the pretraining dataset (IN1k versus IN22k) we see an performance improvement for the ViT-H/14 model. We observe a further performance improvement on semantic tasks by training a larger model ViT-G/16 model on ImageNet-22k. The ViT-H/14 is trained for 300 epochs on IN1k and the equivalent of 900 IN1K epochs on IN22k. The ViT-H/16 is trained for the equivalent of 600 IN1k epochs.

1 CUIDADO Tabla 5: ViT-G/16 mejora lo semántico pero **no** las tareas de bajo nivel; los autores lo atribuyen a parches más grandes (16 frente a 14 px). Otra vez: más abstracción no es mejor en todo.



4 EXTRA Figura 5: la eficiencia es un argumento práctico. Se puede pasar rápido.

Figure 5. **Scaling**. Semi-supervised evaluation on ImageNet-1K 1% as a function of pretraining GPU hours. I-JEPA requires less compute than previous methods to achieve strong performance. Compared to MAE and data2vec, I-JEPA obtains a significant speedup by requiring fewer pretraining epochs. Compared to iBOT, which relies on hand-crafted data-augmentations, a huge I-JEPA model (ViT-H/14) requires less compute than their smallest model (ViT-S/16).

learning performance on semantic and low level tasks when increasing the size of the pretraining dataset (IN1K versus IN22K). Transfer learning performance on these conceptually different tasks improves when pretraining on a larger more diverse dataset.

8 APUNTE Lo mismo en datos y modelo: entrenar con ImageNet-22k y modelos más grandes ayuda en lo semántico.

Scaling model size. Table 5 also shows that I-JEPA benefit from larger model size when pretraining on IN22K. Pretraining a ViT-G/16 significantly improves the downstream performances on image classification tasks such as Place205 and INat18 compared to a ViT-H/14 model, but does not improve performance on low-level downstream tasks — the ViT-G/16 uses larger input patches, which can be detrimental for the local prediction tasks.

8. Predictor Visualizations

The role of the predictor in I-JEPA is to take the output of the context encoder and, conditioned on positional mask tokens, to predict the representations of a target black at the location specified by the mask tokens. One natural question is whether the predictor conditioned on the positional mask tokens is learning to correctly capture positional uncertainty in the target. To qualitatively investigate this question, we visualize the outputs of the predictor. We use the following visualization approach to enable the research community to independently reproduce our findings. After pretraining, we freeze the context-encoder and predictor weights, and train a decoder following the RCDM framework [13] to map the average-pool of the predictor outputs back to pixel space. Figure 6 shows decoder outputs for various random seeds. Qualities that are common across samples represent information that is contained in the average-pooled predictor representation. The I-JEPA predictor correctly captures positional uncertainty and produces high-level object parts with the correct pose (e.g., back of the bird and top of the car).

2 APUNTE §8 responde si el predictor captura la **incertidumbre posicional**: no sabe exactamente dónde está cada parte, pero sí qué parte va ahí.

3 CLAVE Truco metodológico: entrenar un decodificador generativo (RCDM) que convierte la salida del predictor en imágenes. Lo que se repite entre muestras está en la representación; lo que varía, no.

5 CLAVE Resultado: el predictor captura partes de objetos con la **pose correcta** (el lomo del pájaro, el techo del auto).

9. Ablations

Predicting in representation space. Table 7 compares low-shot performance on 1% ImageNet-1K using a linear probe when the loss is computed in pixel-space versus representation space. We conjecture that a crucial component of I-JEPA is that the loss is computed entirely in representation space, thereby giving the target encoder the ability to produce abstract prediction targets, for which irrelevant pixel-level details are eliminated. From Table 7, it is clear that predicting in pixel-space leads to a significant degradation in the linear probing performance.

6 PREGUNTA "We conjecture": es una **conjetura**. La tabla 7 muestra que predecir píxeles es mucho peor, pero no dice **por qué** (¿M1, compresión? ¿M3, menos multimodalidad?). Un experimento que lo separaría: predecir píxeles a través de un cuello de botella (C1).

Masking strategy. Table 6 compare our `multi-block` masking with other masking strategies such as `rasterized` masking, where the image is split into four large quadrants, and the goal is to use one quadrant as a context to predict the other three quadrants, and the traditional `block` and `random` masking typically used in reconstruction-based methods. In `block` masking, the target is a single image block and the context is the

7 APUNTE Variantes de la tabla 6. **Rasterized**: un cuadrante predice los otros tres. **Block**: un solo bloque grande. **Random**: parches sueltos, como en MAE.

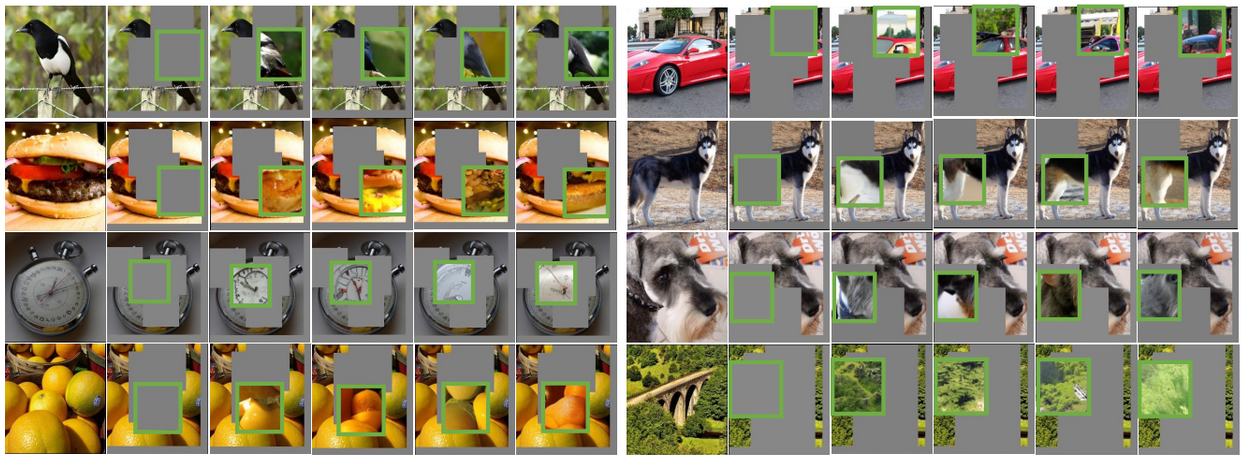


Figure 6. **Visualization of I-JEPA predictor representations**. For each image: first column contains the original image; second column contains the context image, which is processed by a pretrained I-JEPA ViT-H/14 encoder. Green bounding boxes in subsequent columns contain samples from a generative model decoding the output of the pretrained I-JEPA predictor, which is conditioned on positional mask tokens corresponding to the location of the green bounding box. Qualities that are common across samples represent information that contained is in the I-JEPA prediction. The I-JEPA predictor is correctly capturing positional uncertainty and producing high-level object parts with a correct pose (e.g., the back of the bird and top of a car). Qualities that vary across samples represent information that is not contained in the representation. In this case, the I-JEPA predictor discards the precise low-level details as well as background information.

	Targets		Context		
Mask	Type	Freq.	Type	Avg. Ratio*	**Top-1**
`multi-block`	Block$(0.15, 0.2)$	4	Block$(0.85, 1.0)$ $\times$ Complement	0.25	**54.2**
`rasterized`	Quadrant	3	Complement	0.25	15.5
`block`	Block(0.6)	1	Complement	0.4	20.2
`random`	Random(0.6)	1	Complement	0.4	17.6

Avg. Ratio is the average number of patches in the context block relative to the total number of patches in the image.

Table 6. **Ablating masking strategy**. Linear evaluation on ImageNet-1K using only 1% of the available labels after I-JEPA pretraining of a ViT-B/16 for 300 epochs. Comparison of proposed `multi-block` masking strategy. In `rasterized` masking the image is split into four large quadrants; one quadrant is used as a context to predict the other three quadrants. In `block` masking, the target is a single image block and the context is the image complement. In `random` masking, the target is a set of random image patches and the context is the image complement. The proposed `multi-block` masking strategy is helpful for guiding I-JEPA to learn semantic representations.

Targets	**Arch.**	**Epochs**	**Top-1**
Target-Encoder Output	ViT-L/16	500	**66.9**
Pixels	ViT-L/16	800	40.7

Table 7. **Ablating targets**. Linear evaluation on ImageNet-1K using only 1% of the available labels. The semantic level of the I-JEPA representations degrades significantly when the loss is applied in pixel space, rather than representation space, highlighting the importance of the target-encoder during pretraining.

image complement. In `random` masking, the target is a set of random patches and the context is the image complement. Note that there is no overlap between the context and target blocks in all considered strategies. We find `multi-block` masking helpful for guiding I-JEPA to learning semantic representations. Additional ablations on `multi-block` masking can be found in Appendix C.

10. Conclusion

We proposed I-JEPA, a simple and efficient method for learning semantic image representations without relying on hand-crafted data augmentations. We show that by predicting in representation space, I-JEPA converges faster than pixel reconstruction methods and learns representations of a high semantic level. In contrast to view-invariance based methods, I-JEPA highlights a path for learning general representations with joint-embedding architectures, without relying on hand-crafted view augmentations.

2 **PREGUNTA** ¿Es esto un "nivel básico"? El predictor entrega "un pájaro de espaldas en esta posición", no "este pájaro con estas plumas" (C3).

1 **CLAVE** Figura 6: imagen, contexto (la caja verde es la zona a predecir) y muestras decodificadas. **Conserva** forma y pose; **descarta** detalle fino, textura y fondo. Es "lo que se pierde" (ep. §3.7) hecho visible.

3 **CLAVE** **Tabla 6: la máscara lo es todo.** Multibloque 54.2; rasterizado 15.5; bloque 20.2; aleatorio 17.6. Cambiar **qué** se pide predecir cambia la semántica aprendida.

4 **PREGUNTA** Esto es discretizar **la tarea** (ep. tabla 2): los cortes los decide el diseñador. ¿Qué principio explica que ganen varios bloques medianos? El paper no lo dice; solo lo mide.

5 **CLAVE** **Tabla 7:** objetivos = salida del target encoder, 66.9; píxeles, 40.7. Es la diferencia más grande del paper: predecir en espacio abstracto es **el** ingrediente.

6 **CUIDADO** No es una comparación pareada: píxeles con 800 épocas frente a 500. Como píxeles pierde aun con más entrenamiento, la conclusión resiste; pero es un solo modelo y sin barras de error.

7 **CLAVE** Mensaje final: un camino para aprender representaciones generales **sin** aumentos de vista. JEPA-Anything toma esta idea y la lleva a otros "mundos".

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2 **PROFUNDIZAR** LeCun (2022): el programa de investigación detrás de JEPA. Léelo si quieres el porqué de fondo de los modelos del mundo (ep. §2.5).

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1 **PROFUNDIZAR** Rao y Ballard (1999): codificación predictiva en la corteza visual. Es el origen de la idea de "predecir la entrada sensorial".

2 **PROFUNDIZAR** Tian, Chen y Ganguli (2021): por qué los métodos sin negativos (BYOL, SimSiam) no colapsan. Es la teoría que falta en §2.

A. Implementation Details

A.1. Pretraining

Architectures. For I-JEPA pretraining, we use Vision Transformer [29] (ViT) architectures for the context-encoder, target-encoder, and the predictor. While the context-encoders and target-encoders correspond to standard ViT architectures, the predictor is designed as a light-weight (narrow) ViT architecture. Specifically, we fix the embedding dimension of the predictor to 384, while keeping the number of self-attention heads equal to that of the backbone context-encoder. For the smaller ViT-B/16 context-encoder, we set the depth of the predictor to 6. For ViT-L/16, ViT-H/16, and ViT-H/14 context-encoders, we set the depth of the predictor to 12. Finally, the ViT-G/16 uses a predictor of depth 16. I-JEPA is pretrained without a `[cls]` token. We use the target-encoder for evaluation and average pool its output to produce a global image representation.

Optimization. We use AdamW [51] to optimize the context-encoder and predictor weights. Our default batch-size is 2048, and the learning rate is linearly increased from 10^{-4} to 10^{-3} during the first 15 epochs of pretraining, and decayed to 10^{-6} following a cosine schedule thereafter. Following [4, 18], the weight-decay is linearly increased from 0.04 to 0.4 throughout pretraining. The target-encoder weights are identical to the context-encoder weights at initialization, and updated via an exponential moving average thereafter [4, 18, 23, 35, 37, 61]. We use a momentum value of 0.996, and linearly increase this value to 1.0 throughout pretraining, following [4, 18].

Masking. By default, we sample 4 possibly overlapping target blocks masks with random scale in the range $(015, 0.2)$ and aspect ratio in the range $(0.75, 1.5)$. We sample 1 context block mask with random scale in the range $(0.85, 1.0)$ and unit aspect ratio. We subsequently eliminate any regions in the context block mask that overlap with any of the 4 target block masks. The context-block mask and target-block masks are sampled independently for each image in the mini-batch. To ensure efficient batch processing, we restrict the size of all context masks on a co-located GPU to be identical. Similarly, we restrict the size of all target masks on a co-located GPU to be identical. The mask-sampler is efficiently implemented in only a few lines of code in PyTorch [56] using a batch-collator function, which runs in the data loader processes. In short, in each iteration, the data loader returns a mini-batch of images and a set of context and target masks for each image, identifying the patch indices to keep for the context and target views.

A.2. Downstream Tasks

Linear evaluation. When evaluating methods such as iBOT [79], DINO [18] or MAE [36], which leverage Vision Transformers [29] with an additional `[cls]` token, we use the default configurations of VISSL [34] to evaluate all the models on iNaturalist18 [65], CIFAR100 [45], Clevr/Count [42, 75], Clevr/Dist [42, 75], and Places205 [78]. We freeze the encoder and return the best number among the following representations: 1) the `[cls]` token representation of the last layer, 2) the concatenation of the last 4 layers of the `[cls]` token. For each representation, we try two different heads: 1) a linear head, or 2) a linear head preceded by a batch normalization, and return the best number. We use the default data augmentations of VISSL [34]: random resize cropping and horizontal flipping, with the exception of Clevr/Count and Clevr/Dist, where we only use center crop and horizontal flipping, as random cropping interferes with the capability of counting objects and estimating distance, removing critical objects from the scene. For CIFAR100, we resize the images to 224×224 pixels, so as to keep the number of patches equal to that used during pretraining.

Because our I-JEPA implementation uses Vision Transformer architectures without a `[cls]` token, we adapt the default VISSL evaluation recipe to utilize the average-pooled patch representation instead of the `[cls]` token. We therefore report the best linear evaluation number among the following representations: 1) the average-pooled patch representation of the last layer, 2) the concatenation of the last 4 layers of the average-pooled patch representations. We otherwise keep the linear-probing recipe identical.

ImageNet evaluations. To evaluate the I-JEPA on ImageNet [60], we adapt the VISSL recipe to use average pooled representations instead of the `[cls]` token. Following MAE [36], we use the LARS [72] optimizer with a batch-size of 16384, and train the linear probe for 50 epochs. We use a learning rate with a step-wise decay, dividing it by a factor of 10 every 15 epochs, and sweep three different reference learning rates $[0.01, 0.05, 0.001]$, and two weight decay values $[0.0005, 0.0]$.

1 PREGUNTA El predictor es **a propósito** estrecho: 384 canales, con encoders de 768 a 1280. Es un cuello de botella de diseño (ver la tabla 14 y C1).

2 APUNTE Para evaluar se usa el **target encoder** con el promedio de los parches; no hay token [cls].

3 EXTRA Momentum del EMA de 0.996 a 1.0: al final del entrenamiento, el target encoder casi se congela.

4 EXTRA Detalles de implementación de la máscara: se pueden saltar.

5 EXTRA Protocolos de evaluación: solo hacen falta si quieres replicar los números.

Low-shot evaluation. To evaluate our model on the ImageNet-1% low-shot task, we adapt the fine-tuning protocol of MAE [36].We fine-tune our ViT-L/H models for 50 epochs on ImageNet-1% with the AdamW optimizer and a cosine learning rate scheduler. We use a batch size of 512, a learning rate layer decay of 0.75 and 0.1 label smoothing. We use the default randaugment data-augmentations as in MAE. In contrast to the fine-tuning done with MAE, we do not use mixup, cutmix, random erasing or drop path. For the I-JEPA, we use a learning rate /weight decay of $3e^{-5}/5e^{-2}$ for the ViT-L/16, $3e^{-5}/4e^{-1}$ for the ViT-H/14 and $3e^{-5}/4e^{-1}$ for the ViT-H/16_{448}. Similar fine-tuning strategy for low-shot learning has been explored by Semi-VIT in the context of semi-supervised learning [16].

B. Broader Related Work

Self-supervised learning of visual representations with joint-embedding architectures is an active line of research [3, 10, 12, 18, 23, 24, 35, 37, 54, 69, 79]. These approaches train a pair of encoders to output similar embeddings for two or more views of the same image. To avoid pathological solution, many popular joint-embedding approaches use explicit regularization [5, 10, 18, 20] or architectural constraints [24, 35]. Collapse-prevention based on architectural constraints leverage specific network design choices to avoid collapse, for example, by stopping the gradient flow in one of the joint-embedding branches [20], using a momentum encoder in one of the joint-embedding branches [35], or using an asymmetric prediction head [8, 20, 35]. Recent work [62] attempts to theoretically understand (in certain simplified settings) how joint-embedding methods with architectural constraints avoid representation collapse without explicit regularization.

Typical regularization-based approaches to collapse prevention in joint-embedding architectures try to maximize the volume of space occupied by the representations. This is often motivated through the InfoMax [52] principle. Indeed, a long-standing conviction in unsupervised representation learning is that the resulting representations should be both maximally informative about the inputs, while also satisfying certain simplicity constraints [33,50]. The former objective is often referred to as the information-maximization principle (InfoMax), while the latter is sometimes referred to as the parsimony principle [52]. Such approaches to representation learning have been proposed for decades (e.g., [14]), where, historically, simplicity constraints were enforced by encouraging the learned representations to be sparse, low-dimensional, or disentangled, i.e., the individual dimensions of the representation vector should be statistically independent [33]. Modern approaches enforce the simplicity constraints coupled with InfoMax regularization through self-supervised loss terms [6, 40, 41, 44, 55, 64]. One example is the widespread view-invariance penalty [53], often coupled with with independence [10,74] or low-dimensionality constraints, e.g., by projecting representations on the unit hypersphere [20, 35, 37]. However, despite its proliferation, there have also been many criticisms of the InfoMax principle, especially since it is does not discriminate between different types of information (e.g, noise and semantics) [2]. Indeed, the sets of features we wish the model to capture are not always those with the highest marginal entropy (maximal information content).

Orthogonal to the contributions of invariance-based pretraining, another line of work attempts to learn representations by artificially masking parts of the input and training a network to reconstruct the hidden content [67]. Autoregressive models, and denoising autoencoders in particular, predict clean visual inputs from noisy views [8, 9, 19, 36, 67]. Typically, the goal is to predict missing inputs at a pixel level [29, 36, 70], or at a patch token-level, using a tokenizer [9, 68]. While these works demonstrate impressive scalability, they usually learn features at a low-level of semantic abstraction compared to joint-embedding approaches [4].

More recently, a set of approaches attempt to combine both joint-embedding architectures and reconstruction based approaches [30], wherein they combine an invariance pretraining loss with a patch-level reconstruction loss, as in the iBOT method [79]. Since view-invariance based approaches are typically biased towards learning global image representations, thereby limiting their applicability to other computer vision tasks, the idea is that adding local loss terms can improve performance on other popular tasks in computer vision [11, 26, 32]. The framework of contrastive predictive coding [55] is also closely related to this line of work on local loss terms. In the context of images [39], here the idea is to use a contrastive objective combined with a convolutional network to discriminate between overlapping image patch representations. Specifically, the goal is to encourage the representations of an image patch to be predictable of the image patches directly below it, while pushing away the representations of other patch views. In contrast to that work, the proposed I-JEPA method is non-contrastive and does not seek to discriminate between image patches. Rather, the goal is to predict the representations of various target blocks from a single context block. This is achieved with a Joint-Embedding Predictive Architecture, using a predictor network that is conditioned on positional embeddings corresponding to the location of the target block in the image. Qualitative experiments in Section 8 show that the predictor network in our architecture learns to correctly perform this local-to-local region feature mapping, and learns to correctly capture positional uncertainty in the image.

1 EXTRA Protocolo del 1 %: fine-tuning al estilo de MAE. Se puede saltar.

2 PREGUNTA La tensión de fondo: **InfoMax** (conservar toda la información) frente a restricciones de **simplicidad** (parsimonia). Es el cuello de botella informacional de Tishby dicho de otra forma (ep. §3.1, M1).

3 CLAVE La crítica a InfoMax: no distingue el ruido de la semántica, y lo que queremos capturar no es lo de máxima entropía. Por eso conviene predecir solo lo **predecible**.

4 APUNTE La otra línea, la reconstrucción en píxeles o tokens, escala bien pero aprende rasgos de bajo nivel semántico.

5 EXTRA Comparación con métodos de pérdida local: se puede saltar.

Targets		Context	
Scale	Freq.	Scale	**Top-1**
(0.075, 0.2)	4	(0.85, 1.0)	19.2
(0.1, 0.2)	4	(0.85, 1.0)	39.2
(0.125, 0.2)	4	(0.85, 1.0)	42.4
(0.15, 0.2)	4	(0.85, 1.0)	**54.2**
(0.2, 0.25)	4	(0.85, 1.0)	38.9
(0.2, 0.3)	4	(0.85, 1.0)	33.6

Table 8. **Ablation of the target block size for multi-block masking**. Linear evaluation on 1% ImageNet-1K (using only 1% of the available labels); ablating the multi-block target size during I-JEPA pretraining of a ViT-B/16 for 300 epochs. Predicting larger (semantic) blocks improves the low-shot accuracy as long as the context is sufficiently informative.

Targets		Context	
Scale	Freq.	Scale	**Top-1**
(0.15, 0.2)	4	(0.40, 1.0)	31.2
(0.15, 0.2)	4	(0.65, 1.0)	47.1
(0.15, 0.2)	4	(0.75, 1.0)	49.3
(0.15, 0.2)	4	(0.85, 1.0)	**54.2**

Table 9. **Ablation of the context size for multi-block masking**. Linear evaluation on 1% ImageNet-1K (using only 1% of the available labels); ablating the multi-block target size during I-JEPA pretraining of a ViT-B/16 for 300 epochs. Reducing the multi-block context size degrades the low-shot performance.

Targets		Context	
Scale	Freq.	Scale	**Top-1**
(0.15, 0.2)	1	(0.85, 1.0)	9.0
(0.15, 0.2)	2	(0.85, 1.0)	22.0
(0.15, 0.2)	3	(0.85, 1.0)	48.5
(0.15, 0.2)	4	(0.85, 1.0)	**54.2**

Table 10. **Ablation of the targets number for multi-block masking**. Linear evaluation on 1% ImageNet-1K (using only 1% of the available labels); ablating the multi-block number of targets during I-JEPA pretraining of a ViT-B/16 for 300 epochs. Increasing the number of target blocks improve the low-shot accuracy.

C. Additional Ablations

This section follows the same experimental protocol as Section 9. We report the result of a linear probe with a frozen backbone, trained on the low-shot 1% ImageNet-1K benchmark.

Multiblock masking strategy. We present an extended ablation of the multiblock masking strategy where we change the targets block scale (Table 8), the context scale (Table 9) and the number of target blocks (Table 10). We train a ViT-B/16 for 300 epochs using I-JEPA with various `multi-block` settings and compare performance on the 1% ImageNet-1K benchmark using a linear probe. In short, we find that it is important to predict several relatively large (semantic) target blocks, and to use a sufficiently informative (spatially distributed) context block.

Masking at the output of the target-encoder. An important important design choice in I-JEPA is that the target blocks are obtained by masking the *output* of the target-encoder, not the input. Table 11 shows the effect of this design choice on the semantic level of the learned representations when pretraining a ViT-H/16 using I-JEPA for 300 epochs. In the case where masking is applied to the input, we forward-propagate through the target-encoder once for each target region. Masking the output of the target-encoder during pretraining results in more semantic prediction targets and improves linear probing performance.

2 **PROFUNDIZAR** Experimento natural: **aprender** la escala, por ejemplo adaptando la distribución de escalas según la pérdida. Sería pasar de "el diseñador corta" a "el modelo corta" (ep. §5).

1 **PREGUNTA** **Tabla 8: hay un óptimo intermedio de escala.** (0.075, 0.2) da 19.2; (0.15, 0.2), 54.2; (0.2, 0.3), 33.6. Ni muy chico ni muy grande: es la analogía con el nivel básico de Rosch (ep. §3.8, C3).

3 **CUIDADO** El pico es muy agudo (42.4 → 54.2 → 38.9) con una sola corrida, ViT-B, 300 épocas y sin barras de error. Probablemente la escala óptima depende del dominio y del modelo.

4 **CLAVE** **Tabla 9:** más contexto, mejor (31.2 → 54.2). Aquí no hay óptimo intermedio: la información disponible siempre ayuda.

6 **CLAVE** Con un solo objetivo el método casi no aprende (9.0). Predecir **varias** partes desde un mismo contexto es lo que obliga a representar la escena.

5 **PREGUNTA** **Tabla 10:** más objetivos, mejor (1 → 9.0; 4 → 54.2), y **no se prueba más de 4**. El barrido se detiene justo donde "¿cuántas piezas?" se vuelve interesante (C4).

7 **APUNTE** Resumen de los autores: objetivos grandes (semánticos) y contexto informativo (repartido). Coincide con lo que dice el resumen del paper.

Target Masking	Arch.	Epochs	Top-1
Output	ViT-H/16	300	**67.3**
Input	ViT-H/16	300	56.1

Table 11. **Ablating masking output of target encoder**. Linear evaluation on ImageNet-1K using only 1% of the available labels; ablating the effect of masking the target-encoder output during I-JEPA pretraining of a ViT-H/16 for 300 epochs. Masking the output of the target-encoder during pretraining significantly improves the linear probing performance of the pretrained representations.

Predictor depth. We examine the impact of the predictor depth on the downstream low-shot performance in Table 12. We pretrain a ViT-L/16 for 500 epochs using either a 6-layer predictor network or a 12-layer predictor network. The model pretrained using a deeper predictor shows a significant improvement in downstream low-shot performance compared to the model pretrained with a shallower predictor.

Predictor Depth	Arch.	Epochs	Top-1
6	ViT-L/16	500	64.0
12	ViT-L/16	500	**66.9**

Table 12. **Ablating the predictor depth**. Linear evaluation on ImageNet-1K using only 1% of the available labels; ablating the effect of masking the predictor depth for a ViT-L/16 pretrained for 500 epochs. Increasing the predictor depth leads to significant improvement of the linear probe performance of the pretrained representations.

Weight decay. In Table 13, we evaluate the impact of weight-decay during pretraining. We explore two weight decay strategies: linearly increase the weight-decay from 0.04 to 0.4 or use a fix weight-decay of 0.05. Using a smaller weight decay during pretraining improves the downstream performance on ImageNet-1% when fine-tuning. However, this also leads to a degradation of performance in linear evaluation. In the main paper, we use the first weight decay strategy as it improves the performances in linear evaluation downstream tasks.

Weight Decay	Arch.	Epochs	ImageNet-1%	ImageNet Linear-Eval
$0.04 \rightarrow 0.4$	ViT-L/16	600	69.4	**77.8**
0.05	ViT-L/16	600	**70.7**	76.4

Table 13. **Ablating the pretraining weight-decay**. We compare our default pretraining weight decay strategy where we linearly increase the weight-decay from 0.04 to 0.4 to using a fix weight decay of 0.05. Using a smaller weight-decay during pretraining can improve the fine-tuning performance on ImageNet-1%, However, it also leads to a drop of performance in linear evaluation.

Predictor width. We explore the impact of the predictor width in Table 14. We compare I-JEPA using a ViT-L encoder and a predictor with 386 channels to a similar model using a predictor with 1024 channels. Note that the ViT-L encoder has 1024 channels. Using a bottleneck in the predictor width improves the downstream performance on ImageNet 1%.

Predictor Width	Arch.	Epochs	Top-1
384	ViT-L/16	600	**70.7**
1024	ViT-L/16	600	68.4

Table 14. **Ablating the predictor width.** We reports results on ImageNet-1K 1% using fine-tuning. We compare two predictors having a width of either 384 or 1024. Note the I-JEPA encoder is a ViT-L with 1024 channels. Having a width bottleneck in the predictor improves the downstream performances.

D. Finetuning on the full ImageNet

In this section, we report performance on I-JEPA when fine-tuning on the full ImageNet dataset. We focus on the ViT-H/16_{448} as this architecture achieves state-of-art performance with MAE [36].

1 CLAVE **Tabla 11:** recortar a la **salida** del target encoder da 67.3; a la **entrada**, 56.1. El objetivo útil es un parche **visto en contexto**, no un parche aislado.

2 EXTRA Un predictor más profundo (de 6 a 12 capas) ayuda: 64.0 → 66.9. Profundidad sí, ancho no (tabla 14).

3 CUIDADO Tabla 13: un weight decay fijo mejora el 1 % (70.7 frente a 69.4) pero empeora la evaluación lineal completa (76.4 frente a 77.8). Las cifras dependen de qué se mida.

4 PREGUNTA **Tabla 14:** predictor de 384 canales (con cuello de botella), 70.7; de 1024 (sin cuello), 68.4. **Restringir capacidad ayuda sin discretizar nada.** Es una pista directa para C1: parte de lo que se atribuye a discretizar podría ser capacidad (M1).

5 CUIDADO Errata: el texto dice "386 channels" y la tabla, 384. La diferencia es de 2.3 puntos con una sola corrida.

6 PROFUNDIZAR Experimento para P1: comparar un cuello de botella continuo (ancho, ruido) con uno discreto (VQ) **con la misma capacidad en bits**. Si rinden igual, la ventaja era de compresión, no de discreción (C1).

7 EXTRA Apéndice D (fine-tuning completo): se puede saltar. Dato: 87.1 frente a 87.8 de MAE, con 5.3 veces menos épocas.

We use a fine-tuning protocol similar to MAE. Specifically, we fine-tune our model for 50 epochs using AdamW and a cosine learning rate schedule. The base learning rate is set to 10^{-4} and the batch size to 528. We train using mixup [76] set to 0.8, cutmix [73] set to 1.0, a drop path probability of 0.25 and a weight decay set to 0.04. We also use a layer decay of 0.75. Finally, we use the same rand-augment data-augmentations as MAE,

Table 15 reports the fine-tuning results. I-JEPA achieves 87.1 top-1 accuracy. Its performance is less than 1% away from the best MAE model despite I-JEPA being trained for 5.3 times less epochs than MAE. This result demonstrates that I-JEPA is competitive when fine-tuning on the full ImageNet dataset.

Method	Arch.	Epochs	Top-1
MAE [36]	ViT-H/14_{448}	1600	**87.8**
I-JEPA	ViT-H/16_{448}	300	87.1

Table 15. **Finetuning on the full ImageNet dataset**. I-JEPA achieves competitive performance. I-JEPA is close to MAE approach despite I-JEPA being trained for 5.3 times less epochs than MAE.

1 EXTRA Tabla 15: fine-tuning completo; no aporta a nuestras preguntas.

E. RCDM Visualizations

To visualize the representations of a pretrained neural network in pixel space, we use the RCDM framework [13]. The RCDM framework trains a decoder network h_ω, comprising a generative diffusion model, to reconstruct an image $\boldsymbol{x}$ from the representation vector of that image $\boldsymbol{s}_x$ and a noisy version of that image $\hat{\boldsymbol{x}} := \boldsymbol{x} + \epsilon$, where ϵ is an additive noise vector. Concretely, the decoder objective is to minimize the loss function $\|h_\omega(\hat{\boldsymbol{x}}, \boldsymbol{s}_x) - \epsilon\|$. We train each RCDM network for 300,000 iterations using the default hyperparameters [13]. After training the decoder, one can subsequently feed the representation vector of an unseen test image $\boldsymbol{s}_y$ into the decoder along with various random noise vectors to generate several pixel-level visualizations of the representation, thus providing insight into the features captured in the representations of the pretrained network. Qualities that are common across samples represent information that is contained in the representation. On the other hand, qualities that vary across samples represent information that is not contained in the representations

2 APUNTE Cómo leer las figuras 6 a 8: se fija la representación y se varía el ruido del difusor. Lo **constante** entre muestras está codificado; lo que **varía**, no.

In Figure 6, the visualizations are obtained by feeding the average-pooled output of the predictor, conditioned on a specific target region, into the decoder network, along with various random noise vectors. In Figures 7 and 8, the visualizations are obtained by feeding the average-pooled output of the target-encoder into the decoder network, along with various random noise vectors.

E.1. Encoder Visualization

In Figure 7, we visualize the average-pooled I-JEPA representations at the output of our ViT-H/14 target-encoder. The first column contains the original image, while subsequent columns contain synthetic samples obtained by feeding the average-pooled representation of the image into the decoder along with various random noise vectors. Figure 7 suggests that the I-JEPA target-encoder is able to correctly capture the high-level information regarding objects and their poses, while discarding low-level image details and background information.

3 CLAVE Figura 7 (target encoder): conserva objetos y su pose; descarta el detalle fino y el fondo.

Figure 8 shows similar visualizations, but when using an MSN [4] pretrained ViT-L/7 target-encoder to compute the image representations. The MSN method trains a context- and target-encoder using a Joint-Embedding Architecture to enforce invariance of global image representations to various hand crafted data augmentations and missing patches. While the MSN pretrained network is able to capture high level semantic information about the image in the first column, it also exhibits higher variability in the generated samples, e.g., variability in the object pose, object scale, and number of instances. In short, the MSN pretrained discards much of the local structure in the image, which is in stark contrast to I-JEPA, which retains information about much of the local structure in the input image.

4 PREGUNTA Contraste revelador: MSN (invarianza a aumentos) conserva la clase pero **pierde la pose e incluso el número de objetos**; I-JEPA los conserva. Coincide con la tabla 4 (Clevr/Dist: DINO 53.4, I-JEPA 72.4). Cada objetivo decide qué se descarta (ep. §3.7).

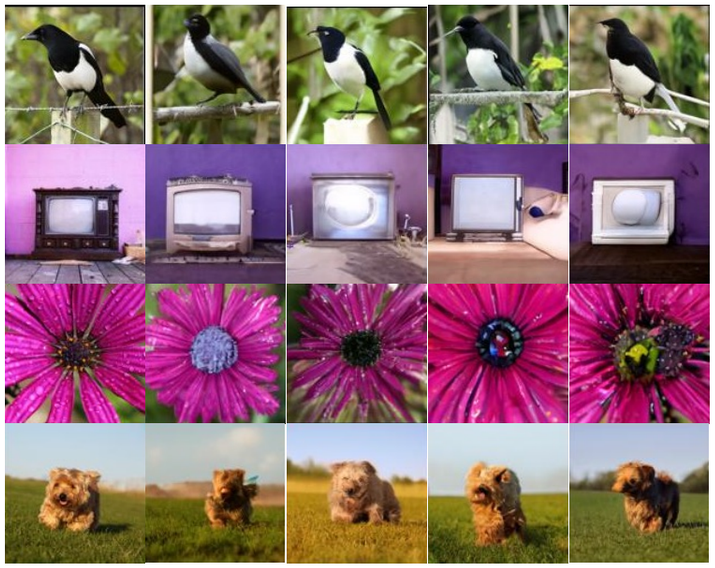


Figure 7. **Visualization of I-JEPA target-encoder representations**. For each image: first column contains the original image; subsequent columns contain samples from a generative model decoding the average-pooled output of a pretrained I-JEPA target-encoder. Qualities that are common across samples represent information that contained is in the I-JEPA representation. I-JEPA is able to correctly capture the high-level information regarding objects and their poses. Qualities that vary across samples represent information that is not contained in the representation. I-JEPA encoder discards the precise low-level details as well as background information.

1

1 CLAVE I-JEPA: las muestras varían en detalles, pero **mantienen** la pose y la posición del objeto.

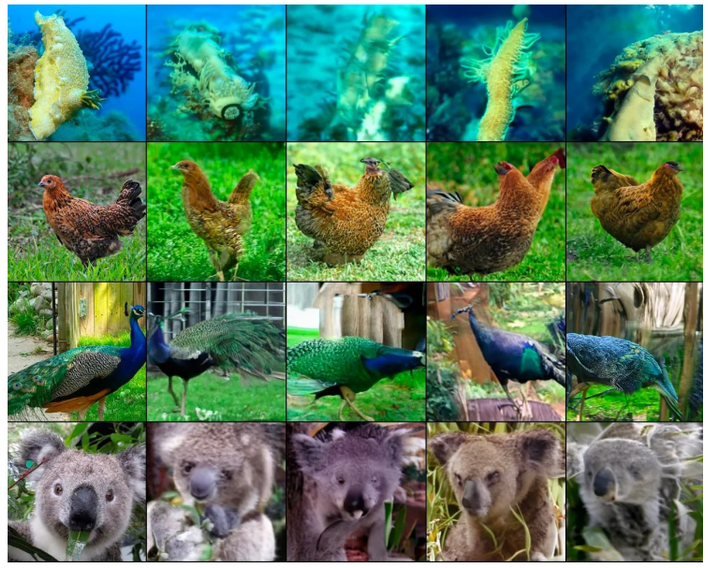


Figure 8. **Visualization of MSN target-encoder representations**. For each image: first column contains the original image; subsequent columns contain samples from a generative model decoding the output of a frozen MSN encoder [4]. Qualities that are common across samples represent information that is contained in the representation. Qualities that vary across samples represent information that is not captured by MSN. Compared to I-JEPA, MSN samples show higher variability. MSN retains less information from the input. In particular, it discards global structure information such as the object pose or even number of instances.

2

2 CUIDADO MSN: las muestras varían incluso en pose y número de objetos. Es evidencia cualitativa con ejemplos elegidos: sirve para la intuición, no como prueba.